

A Map-Level Virtual Human Cell

Integrating 20 measured datasets and 3 machine-learning models into an interactive, perturbable, confidence-scored cell

1. One-line pitch

A single interactive cell where **every human protein** carries its importance, context, wiring, and druggability — and where you can **knock it out, mutate it, overexpress it, infect it, or differentiate it into another cell type** and watch the consequences propagate, each answer backed by a **source and a confidence score**.

2. Executive summary

Biology's knowledge is scattered across dozens of databases and a handful of foundation models, none of which talk to each other. This project fuses **20 measured datasets** and **3 ML models** into one coherent, queryable cell:

- **16,492 localized proteins**, wired by 45,499 signed regulatory + 141,532 physical + 17,432 signaling edges, carrying 12,931 metabolic reactions, 2,039 complexes, drugs, PTMs, disease, and structure.
- **Measured essentiality** for 15,913 genes (DepMap, 1,100 cancer lines) — not predicted.
- **Ensemble confidence** on every prediction, surfacing **1,172 novelty candidates** where measurement, constraint, and model disagree.
- A perturbation/differentiation engine that is **cell-type-specific** and **validated against known biology** (recovers mTOR co-essentiality, EGFR drugs, TP53 targets, HNF4A → liver identity).

The deliverable is a **self-contained interactive HTML cell** + reproducible notebooks. It is a *map*, not a kinetic simulator — it answers "what is connected, what breaks, what's important, in what context, and what to test," not "what is the flux at time t."

3. The problem

- **Fragmentation**: essentiality is in DepMap, regulation in CollecTRI, metabolism in Human-GEM, drugs in DGIdb, structure in AlphaFold — a researcher must query each separately and mentally integrate.
- **Foundation models are siloed and mis-applied**: Geneformer/scGPT are powerful but often benchmarked on the wrong task; their static embeddings score *random* on essentiality, yet are excellent at perturbation.
- **No confidence**: most in-silico predictions arrive with no honest measure of trust.
- **Context blindness**: most "networks" are the same for a neuron and a hepatocyte.

This project addresses all four: integration, right-tool-for-right-job models, ensemble confidence, and cell-type context.

4. Design principles

1. **Map, not simulator**. Structural/logical resolution (what connects to what, what breaks) — kinetics are explicitly out of scope (no clean bulk rate data exists; see §12).
2. **Substrate × Behavior**. Databases supply the *substrate* (facts); models supply the *behavior* (importance, context, dynamics). A complete cell needs both.
3. **Three orthogonal models, each doing only what it's best at** — proven by benchmark, not assumed.
4. **Every label carries provenance and confidence**.

5. Architecture & pipeline stages

```
STAGE 0  DOWNLOAD          20 datasets -> MyDrive/virtual_cell_data/  (download_all_data.ipynb)
STAGE 1  BUILD SUBSTRATE  parse DBs -> localized proteins + all measured layers
STAGE 2  MODEL INFERENCE M1 (essentiality) · M2 (atlas expression) · M3 (Geneformer)  [GPU]
```

STAGE 3 REASONING/FUSE DepMap essentiality + co-essentiality + SL + ensemble confidence + novelty
 STAGE 4 RENDER/INTERACT self-contained cell_complete.html + engines

Build code: `build_cell_complete.py` (assembles `cell_complete.json`) → `build_cell_app_complete.py` (renders HTML).
 Orchestrated by `build_complete_cell.ipynb`.

6. DATA APPENDIX — every dataset: source, stage, use

Legend for **Stage**: 0 download · 1 substrate build · 2 model input · 3 reasoning · 4 render.

#	dataset	source (URL host)	license	size	stage	processing	powers (what it does)
1	UniProt subcellular localization	rest.uniprot.org	CC BY 4.0	4 MB	1	parse to 13 compartment buckets	places each protein in its organelle (the spatial cell)
2	CollecTRI / DoRothEA / TRRUST	omnipathdb / raw repo	CC BY	26 MB	1	signed edges (activate/repress)	gene-regulatory network; KO & overexpression cascades
3	STRING physical links v12	stringdb-downloads.org	CC BY 4.0	27 MB	1	ENSP → symbol, score ≥700	protein–protein binding; cascade propagation
4	SIGNOR	signor.uniroma2.it	CC BY-SA	20 MB	1	directed signed signaling	receptor → kinase → TF signal flow
5	Human-GEM	github SysBioChalmers	CC BY 4.0	1.8 MB	1	ENSG → symbol; reaction per enzyme	genome-scale metabolism (2,549 enzymes)
6	Reactome (Ensembl2Reactome)	reactome.org	CC0	51 MB	1	pathway per gene + cell-cycle phase	pathway modules; cell-cycle tagging
7	EBI Complex Portal (9606)	ftp.ebi.ac.uk	CC0	5 MB	1	UniProt → symbol; members	named complexes (ribosome, TP53–MDM2...)
8	DGIdb interactions	dgidb.org	open (MIT-ish)	12 MB	1	gene → drug → action	drug targeting per gene
9	UniProt PTM features	rest.uniprot.org	CC BY 4.0	5.4 MB	1	count + type of MOD_RES	post-translational modifications
10	CellPhoneDB	github ventolab	MIT	0.5 MB	1	UniProt → symbol pairs	ligand–receptor / cell–cell
11	NCBI gene_info	ftp.ncbi.nlm.nih.gov	public domain	4.9 MB	1	entrez → symbol, locus	genome position; HIV host mapping
12	UCSC CpG islands	hgdownload.soe.ucsc.edu	free	0.7 MB	1	promoter CpG flag	regulatory-element annotation
13	HPO / Open Targets / ClinVar	HPO, OT GraphQL	CC BY	20 MB	1	disease + druggability	disease associations
14	gnomAD constraint (LOEUF)	gnomAD (baked in CSV)	free	—	1,3	LOEUF per gene	germline importance; confidence input
15	AlphaFold / ESMFold	AlphaFold DB / API	CC BY 4.0	on-demand	1	WT vs mutant fold RMSD	mutation structural impact
16	NCBI HIV-1 interaction DB	ftp.ncbi.nlm.nih.gov	public domain	4.8 MB	1	viral → host map	HIV infection engine
17	DepMap CRISPRGeneEffect 24Q2	figshare	CC BY 4.0	382 MB	2,3	Chronos matrix reductions	measured essentiality + co-essentiality + SL
18	HiCCUPS 3D loops (GSE63525)	ftp.ncbi (GEO)	public	0.6 MB	1	TSS → anchor mapping	enhancer-promoter chromatin loops
19	Tabula Sapiens (CELLxGENE)	census S3	CC BY 4.0	streamed	2	per-cell-type mean expression	abundance, cell-type wiring, differentiation
20	Geneformer weights	HuggingFace	Apache-2.0	~GB	2	gene embeddings / perturbation	dark-gene function; cascade enrichment
—	RNAcentral (ncRNA)	ftp.ebi.ac.uk	CC0	160 MB	(pending)	id-mapping	ncRNA inventory (needs targets)
—	ARCHS4 / GEO	maayanlab	CC BY	57.6 GB	(pending)	per-condition expression	disease/condition states
—	Hart CEGv2 / NEGv1	repo	free	tiny	2	truth labels	Model-1 training set

7. THE THREE MODELS — functioning & stats

M1 — Integrated essentiality model · axis: *importance*

- **What:** a gradient-boosted classifier over 14 biological features (LOEUF, TF/regulon, PPI degree, pathways, CpG/enhancers, disease).
- **Data / stage:** trained at Stage 2 on features + DepMap/Hart truth.
- **Stat: cross-validated AUC 0.974** (genome-wide constraint alone: 0.86).
- **Role:** fills essentiality for genes with no CRISPR label (now only 531 gaps, since DepMap covers the rest).

M2 — Atlas / cell-type layer · axis: *context*

- **What:** per-cell-type mean expression from Tabula Sapiens; master TFs by specificity.
- **Data / stage:** Stage 2, streamed + subsampled from the CELLxGENE census.
- **Stat / validation:** recovers canonical master TFs — HNF4A → hepatocyte, GATA4/TBX5 → cardiac, SPI1/CEBPB → macrophage, EOMES → NK, ERG/FLI1 → endothelial.
- **Role:** abundance (node size), which genes are active per cell type, **cell-type-specific network pruning**, and the **differentiation engine** (ON/OFF genes + driver TFs between states).

M3 — Geneformer · axis: *dynamics*

- **What:** pretrained rank-encoding transformer; gene embeddings + in-silico perturbation.
- **Data / stage:** Stage 2, weights from HuggingFace + tokenized atlas cells.
- **Stat (honest):** static embeddings score **AUC 0.53 (random) on essentiality** — which is *why we do not use it for importance*. Its genuine strength is perturbation direction and functional proximity.
- **Role:** enriches the KO cascade with predicted downstream genes; predicts **dark-gene function** by embedding neighbors.

DepMap — measured ground truth (not a model)

- 1,100 cancer cell lines × 18,443 genes of CRISPR Chronos scores.
- **15,913 genes measured**, 1,591 common-essential; used as top-priority truth + co-essentiality + SL.

Design lesson (benchmarked, not assumed): graph nets *lost* to a plain MLP (GCN 0.947 / GAT 0.943 vs MLP 0.973); Geneformer embeddings are random for essentiality; FBA covers only ~17% of human essentiality. So each model is assigned the one job it measurably wins at.

8. THE REASONING / FUSION LAYER

derived signal	method	result
Measured essentiality	fraction of DepMap lines dependent (<-0.5)	15,913 genes, measured
Co-essentiality	correlate every gene's KO profile across 1,100 lines	13,451 genes → functional partners
Synthetic lethality	co-dependent and both individually dispensable	1,256 candidate pairs
Ensemble confidence	agreement across DepMap + LOEUF + M1	15,006 scored
Novelty flags	measurement-vs-constraint disagreement	1,172 candidates (e.g. cancer-dependency)

This layer is where the models *complete each other*: agreement → confidence, disagreement → discovery, embeddings → dark-gene function, co-dependency → combinations.

9. THE CELL MODEL — what each part does

- **Spatial scaffold:** proteins drawn in their real compartments (nucleus, cytoplasm, ER, Golgi, mitochondria, membrane, ...), node size = abundance, ring = disease, color = process.
- **Per-gene identity:** every protein carries — compartment, process, genome locus, essentiality (measured/predicted + % cancer lines), LOEUF, confidence badge, novelty flag, abundance, PTMs, complex membership, signaling partners, drugs, disease, ligand/receptor role, co-essential partners, 3D loops, and its metabolic reactions.
- **Networks:** signed regulation, physical binding, directed signaling, metabolism — all traversable.
- **Functioning pipeline (per gene):** genome → input (regulators) → expressed → function (its reaction, or the genes it switches, or its complex) → output. Every link clickable.

10. THE ENGINES — what you can do

engine	mechanism	output
Remove / Mutate	2-hop cascade over reg+PPI (cell-type-gated) + M3 enrichment; DepMap/constraint lethality	damaged cell: affected proteins, disrupted processes/pathways, viable/inviable
Overexpress	propagate signed regulation	up/down targets, shifted processes
Double knockout	union cascades + shared targets + DepMap co-dependency r	synthetic-lethal flag (measured)
Differentiation	ON/OFF genes between two cell-type states + driver TFs	reprogramming plan
Infect: HIV	host-interaction map + dependency factors	hijacked machinery + weak points
Trace / pipeline / pathway	graph traversal	full mechanistic path, module views

11. END PRODUCT

1. **cell_complete.html** — a self-contained ~10 MB interactive cell (open in any browser or serve at `localhost:8000/cell`). No backend, no dependencies.
2. **Reproducible notebooks** — `download_all_data.ipynb` (data → Drive), `build_complete_cell.ipynb` (full GPU build with all models), `serve_cell.py` (localhost).
3. **A validation report** (from the build) proving it recovers known biology.
4. **A ranked novelty table** — dark-gene functions, cancer-specific dependencies, novel SL pairs — the experiment shortlist.

12. QUESTIONS THE MODEL CAN ANSWER

Importance & essentiality - "Is gene X essential — measured, and in what fraction of cancers?" - "Which genes are essential in cancer but tolerated in the germline?" (novelty flags)

Perturbation & therapy - "What breaks if I knock out X in a hepatocyte vs a neuron?" - "What happens if I overexpress this TF?" - "What is synthetic-lethal with gene X?" (measured co-dependency) - "Which damaged nodes in this cascade are druggable, and by what drug?"

Variants & disease - "This mutation — where does it sit, does it break the fold, what disease?" - "This non-coding variant — which gene does its enhancer loop to?" (3D loops) - "What rewires in disease Y?" (with GEO conditions, pending)

Identity & differentiation - "What master TFs define this cell type?" - "To convert cell A → B, which genes turn on/off and which TFs drive it?"

Function discovery - "This unannotated (dark) gene — what's its likely function, and on what evidence?" - "What proteins act together as a module?" (co-essentiality / complexes)

Infection - "What does HIV hijack, and what are its host-dependency weak points (drug targets)?"

Trust - "How confident is this prediction?" — every answer carries model-agreement confidence + sources.

13. VALIDATION — how we know it works

The build emits a report showing recovery of known biology: - Essentiality set $\approx$ DepMap common-essential (POLR2A/RPL* essential; A2M dispensable). - Co-essentiality recovers real modules — TSC1 ↔ TSC2 ($r=0.91$), the SAGA complex, mTOR/GATOR. - Cell-type masters recover textbook identities (HNF4A → liver, etc.). - Drugs: EGFR → gefitinib/osimertinib/cetuximab; BRAF → vemurafenib. - Perturbation: POLR2A KO → inviable; tissue-restricted genes → tissue-specific lethality.

14. NOVELTY — what new it produces

Novel, evidence-backed **hypotheses** (not proven results), concentrated in three places: 1. **Dark-gene function** (~5,000 genes) via embedding + co-expression + compartment + network. 2. **Model-disagreement dependencies** (1,172 flags) — context/cancer-specific biology the simple rules miss. 3. **Untested synthetic-lethal / regulatory combinations** from co-dependency and 3D loops. Ranked by ensemble confidence → a validation shortlist of hundreds of leads.

15. HONEST LIMITATIONS

- **Not a kinetic simulator** — no rates, concentrations, flux, or time-courses (no clean bulk kcat; BRENDA gated).
- **Bias:** DepMap = cancer lines; Tabula Sapiens = healthy adult — context claims inherit these.

- **Predictions are hypotheses**, requiring experimental validation.
- **Correlation $\neq$ causation** for co-essentiality/co-expression — the confidence layer is the guardrail.
- **Pending**: GEO conditions, RNAcentral targets, full methylation tracks, lipids/ions, tissue/spatial.

16. ROADMAP

- **Now**: full A100 build → real ~40 cell types (M2) + Geneformer enrichment (M3) + validation report.
- **Next**: GEO/condition integration (disease states); real curated SL from DepMap; miRNA targets for RNAcentral.
- **Later**: a grounded **LLM agent** over the cell + models as tools — natural-language, multi-hop, cited answers with confidence. Kinetics only if an open bulk source appears.

17. WHO IT'S FOR

- **Pharma / biotech**: target triage, synthetic-lethal combinations, druggability in context, variant interpretation.
- **Researchers**: hypothesis generation for dark genes and context-specific dependencies.
- **Educators / communicators**: an explorable, mechanistic human cell.

The value proposition, stated plainly: **it turns 20 disconnected datasets and 3 models into one interactive cell that answers importance/context/wiring/perturbation/drug/variant questions across cell types — each with a source and a confidence — and hands you a ranked shortlist of testable, non-obvious leads.** That is exactly what an integrative map should do, and it is honest about being a map, not a simulator.